

Test 15: AB-cloud Hamiltonian construction

Verdict: PASS | Generated: 2026-09-14 23:57:32 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Mixed: Hermiticity/?_TRB are EXACT (machine-?), ?r? check is STATISTICAL.

METHOD: Mixed: Hermiticity/?_TRB are EXACT (machine-?), ?r? check is STATISTICAL.

Test 15 HARDCORE: full plaquette scan 96x96 ($\pm$ -quartet), signed flux 4/4, empty 9021/9021, worst empty flux 0.00e+00.

Result Test 15b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

- `plots/plot\_01.png` ? 1600×1000, ABPlotV23 engine (supersampled)

Plots

- `plots/animation.gif` ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 15 --no-two-pass

Independent verification guide

Full computation log in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[23:57:23.362] [+634.390s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 4 vortices, model=:dirac

FULL COMPUTATION LOG

??

CONSOLE CAPTURE

TEST 15 (pass 2, HARDCORE): AB-Cloud construction ? deep validation

Full-lattice plaquette scan of the $\pm$ -quartet (all plaquettes, signed check)

??

? RUNTIME: 1 build + O(N) plaquette scan at 96x96 (N=9216) ? seconds

Lattice: 96x96 open :dirac, ?=0, quartet q= $\pm$ 1.00 ? 9025 plaquettes scanned

Vortex plaquettes (signed): 4/4 correct ?

Empty plaquettes clean (<1e-9): 9021/9021, worst |flux| = 0.00e+00

??

full signed scan PASS ? 4/4 vortex plaquettes, 9021/9021 empty clean

hermiticity PASS ? verify_hermitian on the quartet H

Test 15b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS